Idea generation techniques

Level 6 · Thinking tools · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ use mind maps, brainstorming and Crazy 8s
- ☐ use SCAMPER, random words and worst possible idea
- ☐ think laterally and borrow ideas from nature or other industries

I DO — Worked example

Read how a full-mark answer is built

Brief: reduce the amount of food wasted in your school canteen.

1 Use SCAMPER to generate three different ideas. Name the letter used for each. (3 marks)

Full-mark answer each ✓ is one mark

Substitute: smaller plates so pupils take less food ✓. Eliminate: remove trays so people only carry what they'll eat ✓. Put to another use: leftover bread goes to a community fridge or is made into croutons ✓.

Why marks are not awarded

"Smaller plates, smaller plates in a different colour, smaller bowls."

1 point: ideas are all variations of one idea and no SCAMPER letters are named.

WE DO — Guided practice

Use the starters and word bank to help you

You want ideas for a reusable water bottle that encourages pupils to drink more.

2 Explain how biomimicry could inspire ideas. (2 marks)

Sentence starters

One reason is … because …

Another reason is … so …

Word bank

nature · cactus · linked · Brief · Biomimicry · copying · problems · solves

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which technique asks you to sketch eight ideas in eight minutes?
☐ Mind map ☐ Crazy 8s ☐ SCAMPER ☐ Six hats

b) In SCAMPER, what does the E stand for?
☐ Evaluate ☐ Eliminate ☐ Explore ☐ Expand

c) Velcro was inspired by burrs sticking to a dog. This is an example of...
☐ Reverse brainstorming ☐ Biomimicry ☐ A survey ☐ Converging

d) What is the first rule of brainstorming?
☐ Only good ideas ☐ No judging ☐ Work alone ☐ Pick one quickly

4 Key terms: write the letter of the matching description next to each term.

1. **Crazy 8s**	____	**A** Eight different ideas sketched in eight minutes
2. **SCAMPER**	____	**B** How could we make it worse? Then reverse the answers
3. **Worst possible idea**	____	**C** Force a link between a random word and the problem
4. **Reverse brainstorm**	____	**D** Generate terrible ideas, then flip them into good ones
5. **Random word technique**	____	**E** Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse
6. **Morphological chart**	____	**F** Combine one option for each feature to make new ideas

Your group has run out of ideas for a library redesign.

5 Describe how you would run a "worst possible idea" session. (3 marks)

6 Explain what is meant by ‘Lateral thinking’. (2 marks)

7 Explain what is meant by ‘Biomimicry’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?